

OLAP Dashboard MVP Requirement & Data Feasibility Analysis

Enterprise Platform Assessment for Retail Client (220 Stores Target Scope | 3-Outlet / 3-Month Data Baseline)

1. Executive Summary

This document provides a comprehensive feasibility analysis for building an enterprise Online Analytical Processing (OLAP) system and management reporting dashboard for a retail client operating approximately 220 stores. The assessment evaluates the 14 functional requirements specified in 'Requirements.pdf' strictly against the sample data provided across three representative outlets (Sites 6, 530, 820 / Site 1070) covering a three-month operational period (April, May, June).

[KEY FEASIBILITY FINDINGS] Out of 14 key dashboard requirement modules, 5 modules are FULLY FEASIBLE for MVP delivery, 5 modules are PARTIALLY FEASIBLE (deliverable using heuristic/proxy calculations or requiring attribute normalization), and 4 modules are NOT FEASIBLE due to structural data gaps in the current sample feed (e.g., lack of POS transaction/footfall logs, PO SLA tracking, targets, and historical YOY trends).

Feasibility Status	Module Count	Percentage	Functional Sections Included
Fully Feasible	5	35.7%	Size Curve, Price Ladder, Category Performance, Store Allocation, Merchandise Buying (Core)
Partially Feasible	5	35.7%	Buying Accuracy, Inventory Health, GMROI, Vendor Performance, AI Recommendations
Not Feasible	4	28.6%	Design Performance, Colour Analytics, Sales Dashboard (Granular/LFL), CEO Dashboard (Complete)

2. Baseline Data Architecture & Field Mapping

The evaluation is based on three structured CSV datasets provided by the client representing the monthly cube aggregate and master data:

- april_demo.csv (Cube Data):** Cube aggregate report containing site-level monthly stock movements, sales, transfers, margin, transit, and pending orders. Grain: [Month, Site Code, Barcode].
- items_demo.csv (Item Master):** Item master table containing hierarchy levels (Division, Section, Department), custom dimension fields (CNAME1-6), descriptions (DESC1-3), vendor name (PARTYNAME), barcode (ICODE), cost rate (RATE), maximum retail price (MRP), and stock-in date.

- **location_demo.csv (Site Master):** Store location lookup table mapping ADMSITE_CODE to store names (e.g., VIP, Gariahat, Andul Road, Metro Retail PRO).

3. Detailed Section-by-Section Requirement Feasibility Analysis

Section 1: Merchandise Buying Dashboard

Feasibility Status: **Partially Feasible (Core Metrics Fully Feasible)**

Scope Description: Top/Bottom SKUs, Fast/Slow movers, Dead stock, Sell-through, Weeks of Cover, Months of Inventory, Fill Rate, Freshness, New Launch Performance, Repeat Order Recommendations, Reorder Alerts, Colour/Size/Price/Fabric/Category/Brand/Season performance.

Feasible Features & Data Mappings:

- **Top/Bottom SKUs & Fast/Slow Movers:** Calculated via NET_SALE_QUANTITY, NET_SALE_AMOUNT, and inventory turnover velocity over the 3 months.
- **Sell-through %:** Formula: $\text{NET_SALE_QUANTITY} / (\text{OPENING_QUANTITY} + \text{GOODS_RECEIVE_QUANTITY} + \text{SITE_TRANSFER_IN_QUANTITY})$.
- **Weeks of Cover (WOC) & Months of Inventory (MOI):** Formula: $\text{CLOSING_STOCK_QUANTITY} / (\text{Average Weekly or Monthly Net Sales Quantity})$.
- **Dead Stock:** Identified where CLOSING_STOCK_QUANTITY > 0 and NET_SALE_QUANTITY == 0 across the 3-month window.
- **Reorder Alerts & Repeat Order Recommendations:** Rule-based logic comparing current WOC against target cover threshold (e.g., WOC < 2 weeks).
- **Hierarchy Performance (Category, Department, Section, Division):** Direct grouping via items_demo.csv hierarchy fields.
- **Size & Price Performance:** Size mapped from CNAME5; Price mapped via MRP / RATE buckets.

Non-Feasible Features & Identified Gaps:

- **Fill Rate:** PENDING_PO_QUANTITY is available, but initial ordered PO quantity and order creation timestamps are missing to measure complete fill rate.
- **Freshness & New Launch Performance:** STOCKINDATE is present in items_demo.csv, but exact first-sale timestamps and launch targets are absent.
- **Colour / Fabric / Season Breakdown:** Colour, Fabric, and Season are not explicitly standard columns; they reside unparsed in CNAME1-4 or DESC1-3 text strings.

MVP Delivery Strategy: Deploy MVP with core sales velocity, stock cover, sell-through, and product hierarchy breakdowns. Add standardized attribute tagging (Colour, Fabric, Season) and PO headers for Phase 2.

Section 2: Design Performance

Feasibility Status: **Not Feasible (Data Structure Gap)**

Scope Description: Design-wise sales by size, sell-through, ranking, GM%, repeat recommendation, cannibalisation, lifecycle, fashion score.

Feasible Features & Data Mappings:

- **Design Sales by Size, Sell-through, Ranking, GM%:** Basic aggregations by ICODE/Style and CNAME5 size tags are possible.

Non-Feasible Features & Identified Gaps:

- **Cannibalisation:** Requires granular daily bill-level cross-basket correlation and style substitution analysis over 12+ months.
- **Design Lifecycle Tracking:** Requires multi-season history; 3 months is insufficient to establish product lifecycle curve.
- **Fashion Score:** No aesthetic, visual, or trend rating attribute dataset exists in the client schema.

MVP Delivery Strategy: Exclude Cannibalisation and Fashion Score from MVP. Provide basic style-level sales and GM% ranking as a interim 'Style Performance' view.

Section 3: Buying Accuracy

Feasibility Status: Partially Feasible (Proxy Metrics)

Scope Description: Bought vs Sold vs Unsold, Markdown, Lost Sales, Overbuy/Underbuy, Buying Accuracy %.

Feasible Features & Data Mappings:

- **Bought vs Sold vs Unsold:** Directly mapped using GOODS_RECEIVE_QUANTITY (Bought), NET_SALE_QUANTITY (Sold), and CLOSING_STOCK_QUANTITY (Unsold).
- **Markdown / Discount Analysis:** Tracked via SALE_DISCOUNT_AMOUNT, SALE_PROMO_AMOUNT, and variance between MRP and RSP.
- **Overbuy / Underbuy:** Calculated by comparing received stock against realized 3-month sales velocity.

Non-Feasible Features & Identified Gaps:

- **Lost Sales:** True lost sales require daily out-of-stock tracking combined with uncaptured demand/footfall, which is not captured in monthly aggregates.

MVP Delivery Strategy: Build Bought vs Sold vs Unsold and Markdown analysis into MVP. Estimate lost sales using stock-out days proxy if stock reaches zero.

Section 4: Store Allocation

Feasibility Status: Fully Feasible

Scope Description: Overstock/Understock, stock-out risk, transfers, replenishment, suggested allocations.

Feasible Features & Data Mappings:

- **Overstock / Understock Identification:** Calculated by comparing store-level WOC against average chain-wide sales velocity across the 3 sites.
- **Stock-Out Risk:** Flagged when CLOSING_STOCK_QUANTITY < projected 14-day sales volume.
- **Transfers & Replenishment Tracking:** Directly tracked via SITE_TRANSFER_IN_QUANTITY, SITE_TRANSFER_OUT_QUANTITY, and WAREHOUSE_TRANSFER_IN/OUT fields.
- **Suggested Allocations:** Heuristic transfer optimization algorithm rebalancing inventory from high WOC stores (Overstocked) to low WOC stores (Understocked).

MVP Delivery Strategy: Fully build Store Allocation dashboard and inter-store rebalancing recommendation engine in Phase 1 MVP.

Section 5: Size Curve Analytics

Feasibility Status: **Fully Feasible**

Scope Description: Bought vs sold by size with future recommended size ratios.

Feasible Features & Data Mappings:

- **Bought vs Sold by Size:** Mapped by joining items_demo.csv (CNAME5 containing size tags e.g., 28", 30", 32", 34", 36") with april_demo.csv receive and sale quantities.
- **Empirical Size Distribution Ratios:** Calculated as percentage sales breakdown per size within each Department/Category.
- **Recommended Size Ratios:** Generated by projecting historical size contribution percentages onto future purchase orders.

MVP Delivery Strategy: Fully deliver Size Curve analytics in MVP. Require client to enforce clean numerical size tagging in CNAME5.

Section 6: Colour Analytics

Feasibility Status: **Not Feasible (Missing Standard Dimension)**

Scope Description: Sales, sell-through, margin, stock left, state/season preferences.

Feasible Features & Data Mappings:

- **Sales, Sell-through, Margin, Stock Left:** Mathematical logic is identical to standard product aggregations.

Non-Feasible Features & Identified Gaps:

- **Standardized Colour Attribute:** Colour is not a dedicated column in items_demo.csv; it is occasionally buried in text strings (DESC1-3) without standardized codes.
- **State & Season Preferences:** Only 3 stores are provided (insufficient geographic spread for state preference); season tags are absent.

MVP Delivery Strategy: Defer Colour Analytics module until client adds a structured 'Colour_Code' column in the product master schema.

Section 7: Price Ladder Performance

Feasibility Status: **Fully Feasible**

Scope Description: Performance by price point including units, sales, margin.

Feasible Features & Data Mappings:

- **Price Point Bucketing:** Dynamic price banding based on MRP or RATE (e.g., <300, 300-500, 500-1000, >1000).
- **Units, Sales, Margin Analysis:** Summation of NET_SALE_QUANTITY, NET_SALE_AMOUNT, and GP_AMOUNT / ADJUSTED_GP_AMOUNT across price bands.

MVP Delivery Strategy: Fully deliver Price Ladder module in Phase 1 MVP.

Section 8: Inventory Health

Feasibility Status: **Partially Feasible**

Scope Description: Inventory value, ageing, dead stock, liquidation, rotation by store/category/state.

Feasible Features & Data Mappings:

- **Inventory Value:** Directly mapped from CLOSING_STOCK_AMOUNT.
- **Dead Stock & Liquidation Candidates:** Flagged for stock with zero sales over 90 days.
- **Rotation by Store & Category:** Turnover ratio calculated per store and department.

Non-Feasible Features & Identified Gaps:

- **Exact Stock Ageing Buckets:** STOCKINDATE in items_demo.csv is static at the item level rather than batch/GRN level; true FIFO ageing requires GRN batch tracking.
- **State Rotation:** Cannot represent state-level analytics with only 3 sample stores.

MVP Delivery Strategy: Provide store and category inventory value, turnover, and dead stock in MVP. Enhance to batch-level ageing in Phase 2.

Section 9: GMROI (Gross Margin Return on Investment)

Feasibility Status: **Partially Feasible**

Scope Description: GMROI by store, buyer, category, vendor, brand.

Feasible Features & Data Mappings:

- **GMROI Calculation Engine:** Formula: Total Gross Profit (GP_AMOUNT) / Average Inventory Value (CLOSING_STOCK_AMOUNT).
- **GMROI by Store, Category, Vendor, Brand:** Aggregated across ADMSITE_CODE, Department/Division, PARTYNAME, and CNAME fields.

Non-Feasible Features & Identified Gaps:

- **GMROI by Buyer:** No 'Buyer_Name' or 'Buyer_ID' dimension field exists in the item master or transaction data.

MVP Delivery Strategy: Deliver GMROI dashboard for Store, Category, Vendor, and Brand. Exclude Buyer breakdown until Buyer mapping is provided.

Section 10: Vendor Performance

Feasibility Status: **Partially Feasible**

Scope Description: OTIF, delays, quality, returns, lead time, sell-through, vendor score.

Feasible Features & Data Mappings:

- **Vendor Sell-Through & Gross Margin:** Aggregated via PARTYNAME across receive, sale, and stock fields.
- **Vendor Returns & Return Rate %:** Tracked via GOODS_RETURN_QUANTITY and GOODS_RETURN_AMOUNT.

- **Vendor Scorecard (Partial):** Composite score based on sales volume, sell-through, margin, and return %.

Non-Feasible Features & Identified Gaps:

- **OTIF (On-Time In-Full), Delays & Lead Time:** No Purchase Order header tables (PO date, expected arrival date, GRN timestamp) are present in the dataset.

MVP Delivery Strategy: Deliver Vendor Commercial Scorecard (Sell-through, Margin, Returns) in MVP. Integrate PO SLA tracking pipeline in Phase 2 for OTIF and Lead Time.

Section 11: Sales Dashboard

Feasibility Status: **Not Feasible (Granularity & Historical Gap)**

Scope Description: Hourly to yearly sales, LFL growth, targets, ABV, UPT, conversion.

Feasible Features & Data Mappings:

- **Monthly & Period Cumulative Sales:** Tracked via NET_SALE_AMOUNT and NET_SALE_QUANTITY over the 3 months.

Non-Feasible Features & Identified Gaps:

- **Hourly / Daily Sales Breakdown:** Data is pre-aggregated at monthly level per store; no POS transaction timestamps exist.
- **LFL (Like-for-Like) YoY Growth:** Requires prior year historical data (April-June 2025); only 2026 sample data is present.
- **Sales Targets vs Actuals:** No Target dataset or budget tables provided.
- **ABV (Average Basket Value), UPT (Units Per Transaction), Conversion:** Requires POS receipt header data (bill count) and footfall sensor data.

MVP Delivery Strategy: Provide Store Monthly Sales Performance in MVP. Re-label as 'Monthly Sales Summary' and request POS bill logs and footfall integration for Phase 2.

Section 12: Category Performance

Feasibility Status: **Fully Feasible**

Scope Description: Sales, margin, inventory, sell-through, WOC, recommendations.

Feasible Features & Data Mappings:

- **Category Hierarchy Aggregations:** Full multi-tier breakdown by Division -> Section -> Department -> Department Alias.
- **Comprehensive Category Metrics:** Summation and ratio calculations for Sales, Margin %, Closing Stock, Sell-Through %, and WOC.
- **Category Matrix Recommendations:** Actionable matrix categorizing categories into Winners (High Margin, High Sell-Through), Underperformers, and Overstocked.

MVP Delivery Strategy: Fully deliver Category Performance dashboard and recommendation matrix in MVP.

Section 13: CEO Executive Dashboard

Feasibility Status: Partially Feasible (Core KPIs Only)

Scope Description: One-page executive KPIs, top/bottom stores/SKUs, inventory, vendor delays.

Feasible Features & Data Mappings:

- **One-Page Executive Summary KPIs:** Total Chain Sales, Gross Profit %, Total Inventory Value, Overall Sell-Through %, and Average WOC.
- **Top / Bottom Performing Stores & SKUs:** Ranking tables based on net revenue and gross margin.
- **Inventory Health High-Level Cards:** Total stock value, dead stock %, stock turn.

Non-Feasible Features & Identified Gaps:

- **Vendor Delays & Footfall Conversion Cards:** Excluded due to missing PO SLA logs and footfall data.

MVP Delivery Strategy: Deliver a streamlined Executive KPI Dashboard focused on financial, sales, inventory, and store performance.

Section 14: AI Recommendations Engine

Feasibility Status: Partially Feasible (Heuristic Rules Baseline)

Scope Description: Actionable suggestions for reorders, transfers, vendor issues, markdowns, allocation.

Feasible Features & Data Mappings:

- **Rule-Based Reorder Recommendations:** Triggered when SKU WOC < Minimum Buffer Threshold and daily sales velocity > 0.
- **Inter-Store Transfer Optimization:** Algorithm identifying matching surplus (High WOC) and deficit (Low WOC) store pairs.
- **Markdown & Liquidation Alerts:** Triggered for items with zero sales velocity over 60-90 days.
- **Allocation Suggestions:** Proportional stock distribution rules based on store historical contribution %.

Non-Feasible Features & Identified Gaps:

- **Advanced Predictive Machine Learning Models:** 3 months of monthly aggregate data is statistically insufficient to train seasonal ML forecasting models (e.g., Prophet, ARIMA, XGBoost).

MVP Delivery Strategy: Deploy a robust Heuristic Rules Engine for automated alerts in Phase 1 MVP. Transition to predictive ML models after accumulating 12+ months of daily data.

4. Comprehensive Feature Feasibility Matrix

The matrix below consolidates every functional requirement, mapping required data fields against current dataset availability and assigning a Phase 1 MVP delivery status.

Module Name	Sub-Feature / Requirement	Required Data Fields	Current Data Availability	MVP Feasibility
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Merchandise Buying	Top/Bottom SKUs & Movers	NET_SALE_QTY, CLOSING_STOCK_QTY	Available in Cube	FEASIBLE
Merchandise Buying	Sell-Through & WOC	OPENING_QTY, RECEIVE_QTY, SALE_QTY	Available in Cube	FEASIBLE
Merchandise Buying	Fill Rate & Lead Time	PO Order Date, Ordered Qty, Delivery Date	Missing PO SLA Logs	NOT FEASIBLE
Merchandise Buying	Colour/Fabric/Season Performance	Standardized Colour/Fabric/Season Columns	Unstructured Text String	PARTIAL
Design Performance	Sales by Size & GM% Ranking	ICODE, CNAME5, GP_AMOUNT, NET_SALE	Available in Cube/Item	FEASIBLE
Design Performance	Cannibalisation & Fashion Score	POS Basket Logs, Style Attributes	Missing POS/Style Score	NOT FEASIBLE
Buying Accuracy	Bought vs Sold vs Unsold	GOODS_RECEIVE_QTY, NET_SALE, CLOSING	Available in Cube	FEASIBLE
Buying Accuracy	Lost Sales	Daily Out-of-Stock Logs, Demand Rate	Missing Daily Stock Logs	PARTIAL
Store Allocation	Overstock/Understock & Transfers	Store WOC, SITE_TRANSFER_IN/OUT	Available in Cube	FEASIBLE
Store Allocation	Suggested Rebalancing Engine	Multi-Store Inventory & Demand Ratios	Available for 3 Stores	FEASIBLE
Size Curve	Bought vs Sold by Size	CNAME5 (Size), RECEIVE_QTY, SALE_QTY	Available in Cube/Item	FEASIBLE
Size Curve	Recommended Size Ratios	Historical Contribution % by Category	Available in Dataset	FEASIBLE
Colour Analytics	Colour Preference & Margin	Colour Code Column, Geographic Store Spread	Missing Colour Tag	NOT FEASIBLE
Price Ladder	Performance by Price Band	MRP, RATE, NET_SALE_QTY, GP_AMOUNT	Available in Cube/Item	FEASIBLE
Inventory Health	Inventory Value & Dead Stock	CLOSING_STOCK_AMOUNT, STOCKINDATE	Available in Cube/Item	FEASIBLE
Inventory Health	FIFO Batch Ageing	GRN Receipt Batch Timestamps	Static Item StockInDate	PARTIAL
GMROI	GMROI by Store/Category/Vendor	GP_AMOUNT, CLOSING_STOCK_AMOUNT	Available in Cube/Item	FEASIBLE
GMROI	GMROI by Buyer	Buyer ID / Buyer Name Dimension	Missing Buyer Tag	NOT FEASIBLE
Vendor Performance	Vendor Sell-Through & Returns	PARTYNAME, RECEIVE_QTY, RETURN_QTY	Available in Cube/Item	FEASIBLE
Vendor Performance	OTIF & Delays	PO Header, Promised Date, Actual GRN Date	Missing PO Header Table	NOT FEASIBLE

Sales Dashboard	Monthly Revenue & Gross Margin	NET_SALE_AMOUNT, GP_AMOUNT, ADMSITE	Available in Cube	FEASIBLE
Sales Dashboard	Hourly Sales, LFL YoY, ABV/UPT	POS Bill Logs, Prior Year Data, Footfall	Missing POS & Historical	NOT FEASIBLE
Category Performance	Category Hierarchy Metrics & Matrix	Division, Section, Department, WOC	Available in Cube/Item	FEASIBLE
CEO Dashboard	Executive High-Level KPIs	Total Revenue, GP %, Stock Value, Top Stores	Available in Cube	FEASIBLE
AI Recommendations	Heuristic Reorders, Transfers & Markdowns	Rule Engine (WOC, Velocity, Dead Stock)	Feasible via Rules	FEASIBLE
AI Recommendations	Predictive Machine Learning Forecasting	Multi-Year Daily Time Series Data	3-Month Scope Limit	PARTIAL

5. Phase 1 MVP Scope & Phase 2 Data Architecture Roadmap

To ensure a successful MVP launch for the client while laying the groundwork for scaling across all 220 stores, we recommend a phased implementation strategy:

Phase 1: Deliverable MVP Scope (Immediate Baseline)

- **Core Modules:** Store Allocation & Inventory Rebalancing Engine (Covering 3 sample outlets with rebalancing rules).
- **Hierarchy Views:** Category, Department, Section, and Division Multi-Level Performance Dashboard.
- **Merchandise Tools:** Size Curve Analytics & Recommended Size Ratio Calculation.
- **Pricing Analysis:** Price Ladder Performance & Price Band Elasticity Views.
- **Vendor Views:** Vendor Commercial Scorecard (Sell-Through %, Gross Profit Contribution, Goods Return Rate %).
- **Stock Health:** Inventory Health & Dead Stock Liquidation Candidate Engine (3-Month zero-sales velocity).
- **Financial Metrics:** GMROI Engine by Store, Category, Vendor, and Brand.
- **Action Engine:** Heuristic AI Recommendation Center for Reorders, Store Transfers, and Stock Markdowns.

Phase 2: Required Data Pipeline Enhancements for 220-Store Scale

- **1. POS Transaction Header Feed:** Integrate daily POS receipt header and item line tables (capturing Basket ID, Receipt Timestamp, Units Per Basket, Bill Amount) to enable ABV, UPT, Hourly Sales, and Cannibalisation analysis.
- **2. Purchase Order & Supply Chain SLA Feed:** Ingest Purchase Order creation tables, promised delivery dates, and GRN receipts to compute true OTIF, Delivery Lead Time, and Vendor SLA Scores.
- **3. Multi-Year Historical Series:** Ingest prior year historical data (2024-2025) to enable Like-for-Like (LFL) YoY growth calculations and train predictive ML seasonal demand models.
- **4. Footfall & Target Master Tables:** Incorporate store footfall sensor counts and monthly store sales targets to unlock Conversion Rate % and Target vs Actual achievement KPIs.
- **5. Master Data Attribute Enrichment:** Enforce standard attribute tagging in Product Master for Colour_Code, Fabric_Type, Season_Code, and Buyer_ID.